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<title>TradeMatrix – Project Description</title>
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</style>
</head>
<body>
  <h1>TradeMatrix: Project & Research Specification</h1>
  <div class="box">
    <strong>Core Objective:</strong> Build an institutional-grade financial ML pipeline while scientifically investigating whether an Attention-augmented LSTM architecture provides genuine trading edge after zero-leakage validation and market friction (0.05% fees + slippage).
  </div>
  <h2>1. Research Hypotheses</h2>
  <ul>
    <li><strong>Primary Research Question:</strong> Does augmenting an LSTM recurrent network with Multi-Head Self-Attention improve out-of-sample directional return forecasting and risk-adjusted portfolio performance compared to a baseline LSTM and Buy-and-Hold benchmark?</li>
    <li><strong>Hypothesis 1 (Stationarity):</strong> Predicting stationary log returns ( $\ln(P_t / P_{t-1})$ ) eliminates the artificial accuracy artifacts caused by non-stationary price tracking.</li>
    <li><strong>Hypothesis 2 (Multi-Modal Features):</strong> Technical indicators (RSI, MACD, Rolling Volatility) provide orthogonal momentum and volatility signals superior to raw univariate price series.</li>
    <li><strong>Hypothesis 3 (Economic Realism):</strong> A higher statistical accuracy (RMSE/Directional Hit Rate) does not necessarily translate into positive trading alpha once transaction costs and slippage are deducted.</li>
  </ul>
  <h2>2. System Upgrade Architecture</h2>
  <table>
    <tr><th>Component</th><th>Legacy (GetYourStock)</th><th>TradeMatrix Institutional Upgrade</th></tr>
    <tr><td><strong>Target</strong></td><td>Raw Close Price (Non-stationary)</td><td>Forward 1-Day Log Return & Directional Sign</td></tr>
    <tr><td><strong>Splitting</strong></td><td>Random Shuffle / Lookahead Leakage</td><td></td></tr>
    <tr><td><strong>Model</strong></td><td>Vanilla Single-Layer LSTM</td><td>Bi-LSTM / Uni-LSTM + Multi-Head Self-Attention with Huber Loss</td></tr>
    <tr><td><strong>Metrics</strong></td><td>Statistical Loss only (MSE / R2)</td><td>Annualized Sharpe, Sortino, Calmar, Maximum Drawdown (MDD)</td></tr>
    <tr><td><strong>Friction</strong></td><td>Zero friction assumed</td><td>0.05% Transaction fees + Execution slippage + Threshold Tuning</td></tr>
    <tr><td><strong>Interface</strong></td><td>Jupyter Notebook script</td><td>FastAPI Backend + Streamlit UI</td></tr>
  </table>
</body>
</html>

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